

a6:f5:30:6b:af:6d:05:0a:f2:3b:a4:bd:6b:9d:d9:60.

Group membership is enforced by limiting access to the epoch keys. Only Nodes that possess the input epoch key can derive a given operational key. Lack of possession of a particular epoch key restricts access, based on the distribution policy of the epoch keys.

The following diagram shows the process by which operational keys are derived from the epoch key material:

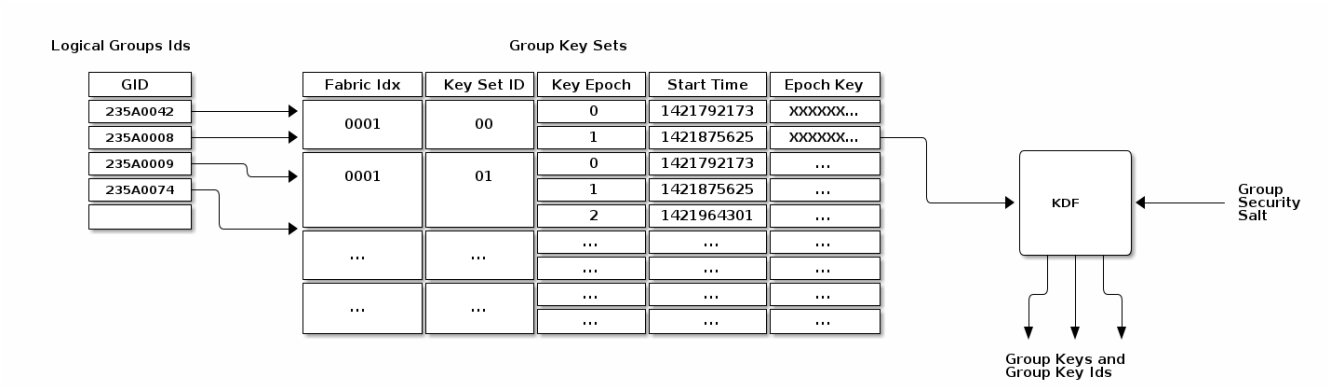


Figure 17. Group Key Derivation

4.17.2.1. Group Security Info

A hard-coded *group security info* is used to diversify the set of operational group keys. This value is hard-coded into the standard’s implementation, and thus is distributed with the associated code. Should the standard’s security mechanisms need to evolve (e.g. to upgrade encryption from AES-128 to AES-256), the *group security info* can be changed to ensure that new keys will be derived for use in the new algorithm. The *group security info* SHALL be the byte stream "GroupKey v1.0", i.e. 0x47 0x72 0x6f 0x75 0x70 0x4b 0x65 0x79 0x20 0x76 0x31 0x2e 0x30.

With the exception of the *group security info*, all input key material SHALL be maintained on a per-Fabric basis.

4.17.2.2. Group Key Set

A *group key set* limits the key derivation process to Nodes within the respective operational groups. Access to a *group key set* is limited based on the functionality provided by a Node and/or the privilege afforded to it. For example, certain home security devices, such as a security system or door lock, may have access to a "Physical Access" *group key set*, while devices such as light bulbs or window coverings would not.

Operational group key lifetime is limited by assigning an expiration time to each *epoch key* in a given *group key set*. By constraining the validity of a given epoch key to an epoch, the ability for members to derive and operate with an operational group key can be constrained to particular periods of time. Epoch keys may be rotated on a periodic basis, and denying access to updated versions of these keys serves as a means to eject group members.

4.17.3. Epoch Keys

Epoch keys provide a means for limiting the lifetime of derived operational group keys. They also